

JKBOSE 2026

CLASS 11TH



Relations and Functions

MATHS

LECTURE-02

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TOPICS TO BE COVERED

1**What is Relation?****2****Exercise Discussion 2.2****3****Related Problems**

Cartesian product of sets:

$$A \times B = \{ (a, b) : a \in A, b \in B \}$$

5

Question



$$a^x = a^y, x=y \quad a \neq 0, -1, 1.$$

If $A = \{x \in \mathbb{Z} : x^2 - 2x + 1 = 0\}$ and $B = \{x : 3^{(x-1)(x-2)(x-3)} = 1\}$, then $A \times B$ is equal to

$$a^0 = 1, a \neq 0$$

$$\frac{(x-1)(x-2)(x-3)}{3} = 3^0$$

~~1~~

$\{(1, 2), (1, 3), (1, 5), (3, 7)\}$.

2

$\{(1, 1), (1, 2), (1, 3)\}$

~~3~~

$\{(1, 2), (1, 7), (1, 6)\}$

~~4~~

$\{(1, 1), (2, 1), (3, 1)\}$

$$\Rightarrow (x-1)(x-2)(x-3) = 0$$

$$\Rightarrow x-1=0, x-2=0, x-3=0.$$

$$\Rightarrow x=1, 2, 3$$

$$\text{So, } B = \{1, 2, 3\}$$

$$\& A = \{1\}$$

$$\rightarrow A \times B = \{(1, 1), (1, 2), (1, 3)\}$$

$$\begin{aligned} a^x &= a^y \\ \Rightarrow x &= y, \text{ provided} \\ a &\neq 0, -1, 1. \end{aligned}$$

$$A = \{x \in \mathbb{R} : x^2 - 2x + 1 = 0\}.$$

$$\text{Here } x^2 - 2x + 1 = 0$$

$$\Rightarrow (x-1)^2 = 0$$

$$\Rightarrow x-1 = 0$$

$$\Rightarrow x = 1$$

$$\Rightarrow A = \{1\}.$$

Question

If $A = \{x \in \mathbb{Z} : -2 \leq x \leq 2\}$ and $B = \{y \in \mathbb{N} : y \leq 3\}$, then the number of Subsets of $A \times B$ is

~~1~~

2^5

~~2~~

2^8

~~3~~

2^{15}

~~4~~

2^{10}

$$A = \{-2, -1, 0, 1, 2\}$$

$$B = \{1, 2, 3\}$$

$$\Rightarrow n(A \times B) = n(A) \times n(B)$$

$$\Rightarrow n(A \times B) = 5 \times 3 = 15$$

$$\text{So, number of Subsets of } A \times B \\ = 2^{15}$$

Relations

• Any Subset R of $A \times B$ is said to be a relation from A to B .

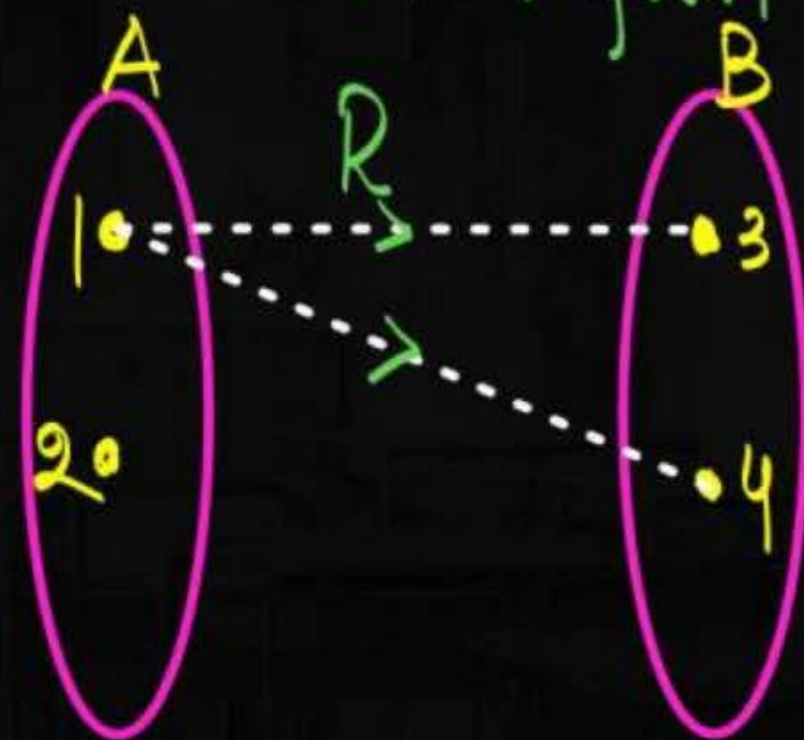
* $\phi \subset A \times B$ (empty relation)
 * $A \times B \subseteq A \times B$ (universal relation) (Trivial Relations)

* $A = \{1, 2\}$ and $B = \{3, 4\}$

$\Rightarrow A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$

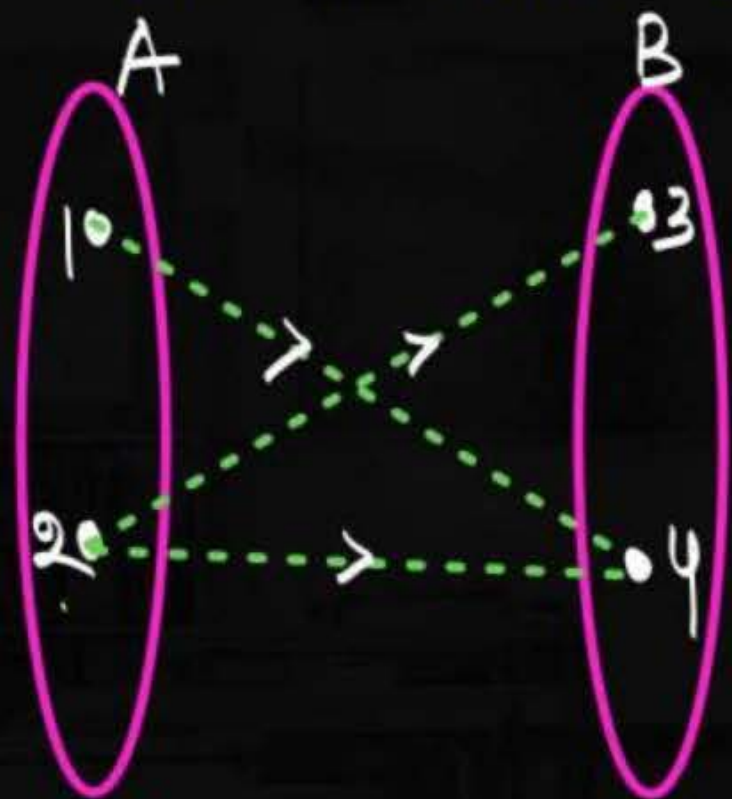
* Here $R = \{(1, 3), (1, 4)\} \subset A \times B$

$\Rightarrow R$ is a relation from A to B .



$$** R = \{(1,4), (2,3), (2,4)\} \subset A \times B$$

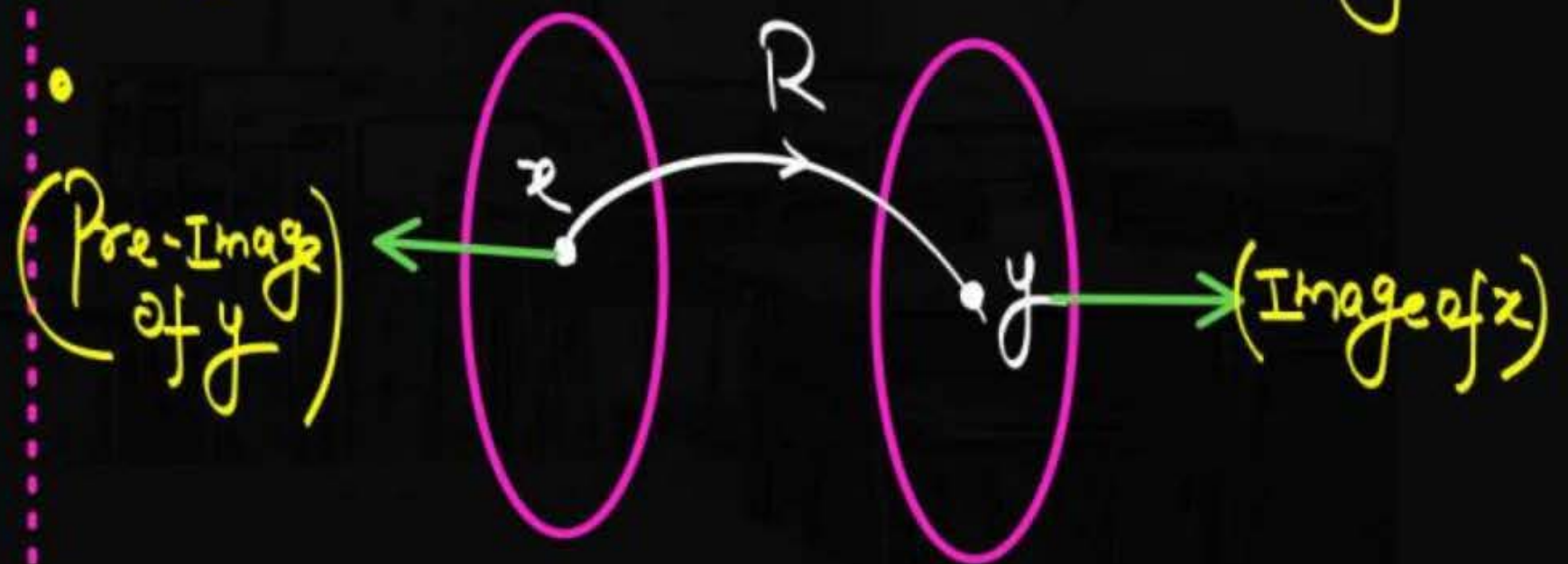
$\Rightarrow R$ is a relation from A to B .



- 4 is the Image of 1 & 2
- 3 is the Image of 2.

Note:

- $A \times B$ is a relation known as **Improper relation**.
- Any other relation is known as **Proper relation**.
- $(x, y) \in R \Leftrightarrow x$ is related to y .
- $(x, y) \notin R \Leftrightarrow x$ is not related to y .



Defination: If R is a Relation A to B , then

- Set of first entrise of R is known as the **Domain of R**
- Set of Second entrises of R is known as the **range of R**
- Set B is known as the **Co-Domain of R**

$$* A = \{1, 2\}, B = \{3, 4\}$$

$$\Rightarrow A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

$$\text{let } R = \{(1, 3), (1, 4)\} \subset A \times B$$

$$\Rightarrow \text{Domain of } R = \{1\}$$

$$\text{Range of } R = \{3, 4\} = B$$

$$\text{Co-Domain of } R = \{3, 4\} = B$$

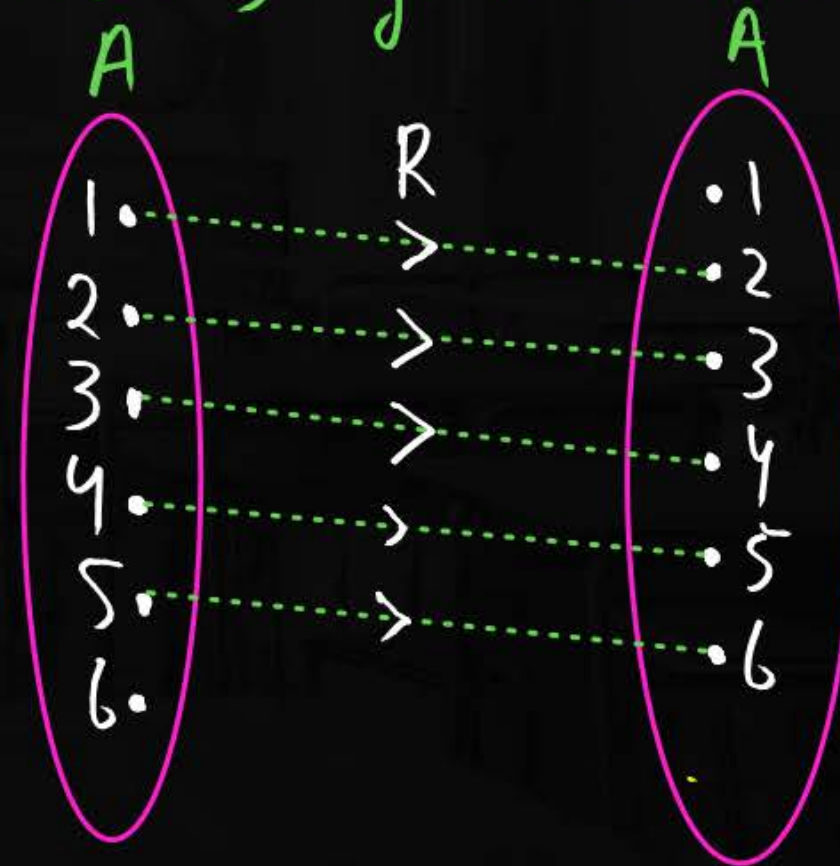
EXAMPLE

Let $A = \{1, 2, 3, 4, 5, 6\}$. Define a relation R from A to A by $R = \{(x, y) : y = x + 1\}$

- Depict this relation using an arrow diagram.
- Write down the domain, codomain and range of R .

Given Relation is,
 $R = \{(x, y) : y = x + 1\} \subset A \times A$
 $\Rightarrow \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$

So, the arrow Diagram is Shown Below:



EXAMPLE

Find the domain and range of the relation R given by

$$R = \{(x, y) : y = x + \frac{6}{x}; \text{ where } x, y \in \mathbb{N} \text{ and } x < 6\}.$$

$\mathbb{N} \rightarrow$ Co-Domain
of R

$$(Sol) \text{ Here } R = \{(x, y) : y = x + \frac{6}{x}, \quad x < 6 \ \& \ x, y \in \mathbb{N}\} \subset \mathbb{N} \times \mathbb{N}$$

$$\Rightarrow R = \{(1, 7), (2, 5), (3, 5)\}$$

$$\text{So Domain of } R = \{1, 2, 3\}$$

$$\text{Range of } R = \{7, 5\}$$

Therefore Domain of $R = \{1, 2, 3, 4, 5\}$

Range of $R = \{2, 3, 4, 5, 6\}$

Co-Domain of $R \{1, 2, 3, 4, 5, 6\} = A$

Q): If $A = \{1, 2\}$ & $B = \{2, 3, 4\}$, then the

number of relations from A to B is,

~~(a) 2^{10}~~ ~~(b) 2^4~~ ~~(c) 2^8~~ ~~(d) 2^4~~ $n(A \times B)$
 $= 2 \times 3 = 6$

$$\begin{aligned} \text{* No of Relations from } A \text{ to } B \\ &= \text{no of Subsets of } A \times B \\ &= 2^{n(A \times B)} \\ &= 2^{n(A) \times n(B)} \end{aligned}$$

Question

Let $n(A) = m$, and $n(B) = n$. Then the total number of non-empty relations that can be defined from A to B is

~~1~~

m^n

~~2~~

$n^m - 1$

~~3~~

$mn - 1$

~~4~~

$2^{mn} - 1$

$$\begin{aligned} & n(A) \times n(B) \\ & \quad \quad \quad 2 \quad - 1 \\ & = 2^{mn} - 1 \end{aligned}$$

Question

Let $A = \{1, 2\}$ and $B = \{3, 4\}$ Then, the number of relations from A to B is

~~1~~ 10

☒ 2 16

~~3~~ 4

~~4~~ 8

$$2 \times 2$$
$$2 = 2 = 16$$

Question

$x \wedge y \Leftrightarrow (x, y) \in R$ co-Primes

A relation R is defined from $\{2, 3, 4, 5\}$ to $\{3, 6, 7, 10\}$ by $xRy \Leftrightarrow x$ is relatively to y .

Then, domain of R is

$$R = \{(x, y) : x \text{ is relatively prime to } y\} \subset A \times B$$

$$\Rightarrow R = \{(x, y) : H(\text{of } (x, y)) = 1\}$$

$$\Rightarrow R = \{(2, 3), (2, 7), (3, 7), (3, 10), (4, 3), (4, 7), (5, 3), (5, 6), (5, 7)\}$$

$$\Rightarrow \text{Domain of } R = \{2, 3, 4, 5\}$$

1

$\{2, 3, 5\}$

2

$\{3, 5\}$

3

$\{2, 3, 4\}$

4

$\{2, 3, 4, 5\}$

Question

$$a^x = a^y, a \neq 0, -1, 1 \Rightarrow a = y$$



If $A = \{x \in \mathbb{Z} : 2^{x^2-3x+2} = 1\}$ and $B = \{y \in \mathbb{N} : 2y + 1 < 11\}$, then the number of relations from A to B is :

~~1~~

$$2^5$$

$$2^{x^2-3x+2} = 1 = 2^0$$

$$\Rightarrow 2^{x^2-3x+2} = 2^0$$

~~2~~

$$2^8$$

$$\Rightarrow x^2 - 3x + 2 = 0$$

$$\Rightarrow x^2 - 2x - x + 2 = 0$$

$$\Rightarrow x(x-2) - 1(x-2) = 0$$

~~3~~

$$2^6$$

$$= (x-1)(x-2) = 0$$

$$\Rightarrow x = 1, 2$$

~~4~~

$$2^{10}$$

$$\text{So } A = \{1, 2\}$$

$$2y + 1 < 11$$

$$\Rightarrow 2y < 11 - 1$$

$$\Rightarrow 2y < 10$$

$$\Rightarrow y < \frac{10}{2} = 5$$

$$\Rightarrow y < 5$$

$$\Rightarrow B = \{y \in \mathbb{N} : y < 5\}$$

$$\Rightarrow B = \{1, 2, 3, 4\}$$

$$\text{So no. of Relations} \\ = 2^{2 \times 4} = 2^8$$

Question(JEE MAINS-2019)

Let Z be the set of integers. If $A = \{x \in Z : 2^{(x+2)(x^2-5x+6)} = 1\}$ and $B = \{x \in Z : -3 < 2x - 1 < 9\}$, then the number of subsets of $A \times B$ is :

1 2^{18}

2 2^{10}

3 2^{15}

4 2^{12}

Home Work

Problem

Thank
You